



AI-Assisted Tumor Segmentation and Margin Distance Estimation in Breast Histopathology Images Using Yolo_seg

A Proof-of-Concept Study

Junior Project Report

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October 2026/2027

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Abstract

Breast-conserving surgery aims to remove a breast tumor together with a margin of healthy tissue. When the removed specimen is later found to have tumor cells at or near its edge, the patient often needs a second operation. Assessing margins during surgery is still difficult: the available methods are slow, limited in coverage, costly, or depend on scarce expert pathologists.

This project is a **proof-of-concept** study that combines artificial intelligence and pathology. We train a YOLO-based instance segmentation model (YOLO-seg) to delineate tumor regions in H&E-stained histopathology images from the BCSS dataset, and we develop an algorithm that converts the segmentation into a quantitative measurement: the shortest distance, in millimetres, between the tumor and the tissue boundary on whole-slide images (TCGA-BRCA). The work is organized over three semesters: (1) segmentation and distance estimation on public data, (2) laboratory experiments on microscope images from the university labs, including domain adaptation by freezing early layers and fine-tuning, and (3) a decision-support layer based on a RAG system or an AI agent.

We state the limitations openly. In particular, the tissue boundary measured on a slide is a proxy and not the inked surgical margin, so the results are a technical demonstration and not clinically interpretable.

Keywords: breast cancer, surgical margin, histopathology, YOLO-seg, instance segmentation, whole-slide images, stain normalization, domain shift.

Part I – Theoretical Background

This part presents the ideas on which the project is based and which a student needs while working on it. It is divided into a medical background and a technical background.

1. Medical Background

1.1 Breast cancer and breast-conserving surgery

The treatment of early-stage breast tumors is based on removing the tumor with a rim of healthy tissue (lumpectomy, or breast-conserving surgery – BCS) instead of removing the whole breast, usually followed by radiotherapy. The most common histological types are:

- **Invasive ductal carcinoma (IDC):** the most frequent type.
- **Invasive lobular carcinoma (ILC).**
- **Ductal carcinoma in situ (DCIS):** a pre-invasive tumor confined to the milk ducts.

1.2 Surgical margins

The margin is the surface of the removed specimen. The surgeon marks it with ink so that the pathologist knows its orientation.

- **Negative margin:** no tumor cells at the ink, or at a sufficient distance from it. **Positive margin:** tumor at the ink or very close to it.
- The accepted distance depends on the tumor type and on the guideline. For invasive carcinoma, commonly used American guidelines require "no tumor on ink"; for DCIS, 2 mm is often used. In this project **2 mm is used as an experimental, adjustable alert threshold**.
- A positive margin increases the risk of local recurrence and often requires a second operation (re-excision).

1.3 Histopathology imaging

- **H&E staining:** hematoxylin stains nuclei blue-purple; eosin stains cytoplasm and stroma pink.
- **Frozen sections:** fast, but of lower quality because of freezing artifacts, and difficult for fatty breast tissue.
- **Formalin-fixed paraffin-embedded (FFPE) tissue:** the highest quality, but it takes hours to days.
- **Whole-slide images (WSI):** high-resolution digital scans of a whole slide, in formats such as SVS, stored as image pyramids. The metadata contains the **MPP** (microns per pixel), which is the basis for converting pixel distances into millimetres.

2. Technical Background

2.1 Deep learning fundamentals

1. **Convolutional neural networks (CNN):** the backbone extracts features, the neck fuses them, and the head produces the outputs.

2. **Computer-vision tasks:** classification, object detection, semantic segmentation and instance segmentation.
3. **The YOLO family:** single-stage detectors. In YOLO-seg models (v8, v10, v11) an additional branch produces a mask for each object (prototype masks combined with per-instance coefficients).
4. **Loss functions:** box loss, class loss, distribution focal loss (DFL) and segmentation loss.
5. **Transfer learning:** starting from weights pre-trained on a large dataset (e.g., COCO) and fine-tuning on our data.
6. **Freezing:** keeping the early layers (generic features such as edges and texture) fixed and re-training only the last layers on new data. This is useful when laboratory data are scarce.
7. **Overfitting and augmentation:** how to recognize them from training and validation curves and how to reduce them.

2.2 Evaluation metrics

- **IoU and Dice:** overlap between the predicted and the ground-truth mask.
- **Precision, recall, mAP@0.5 and mAP@0.5:0.95**, for both boxes and masks.
- In a medical setting **recall** is especially important, because missing a tumor region is more harmful than a false alarm.
- **For the distance measurement:** MAE and RMSE between the computed and the reference distance.

2.3 Digital pathology data processing

1. **Converting masks to polygons** in the YOLO label format.
2. **Tiling** into patches of 512×512 pixels at a fixed resolution.
3. **Patient-level (or slide-level) splitting** to prevent **data leakage**, which gives falsely optimistic results.
4. **Domain shift:** a model trained on data from one source may perform worse on images from another laboratory because of differences in scanner, illumination, staining and magnification.
5. **Stain normalization,** for example the **Macenko** method, to reduce the effect of staining differences.
6. **Resolution matching:** downsampling laboratory images to the resolution used in training.
7. **Reading whole-slide images:** using libraries such as OpenSlide, understanding pyramid levels and extracting MPP from the metadata.

2.4 Computing the tumor-to-boundary distance

1. **Tissue boundary extraction:** for example Otsu thresholding on a grayscale or saturation channel, followed by morphological clean-up.
2. **Tumor boundary extraction** from the segmentation mask.
3. **Shortest distance** between the two boundaries, using a distance transform or contour-to-contour search.

4. **Unit conversion** from pixels to millimetres using the MPP:

```
distance (mm) = distance (px) × MPP (µm/px) ÷ 1000
```

- The main sources of error are the accuracy of the calibration (MPP) and the accuracy of the mask boundaries.

2.5 Real-time processing (later stage)

Capturing a live stream from the microscope camera with OpenCV, reaching at least 25 frames per second, and accelerating inference with ONNX or TensorRT.

2.6 Decision-support layer (final stage)

- **RAG (retrieval-augmented generation):** storing passages from clinical protocols in a vector database (e.g., ChromaDB) and retrieving the relevant ones for each case.
- **AI agent:** an agent that coordinates the results and generates reports.
- The system is a **decision-support aid** and is not a replacement for the surgeon or the pathologist.

Part II – Medical and Scientific Problem

3. The Medical Problem

A considerable proportion of patients who undergo breast-conserving surgery (reported re-excision rates range roughly between 15% and 30%, depending on the centre and the margin definition) are found after the operation to have positive margins and need a second surgery. This causes psychological and surgical burden, additional cost and delays in adjuvant treatment. The reason is that the final margin assessment is made after the operation in the pathology laboratory, and there is no fast, accurate and widely available way to assess margins during surgery:

- **Frozen sections** take time, cover only limited parts of the specimen surface and are of lower quality.
- **Other supporting techniques** (specimen radiography, spectroscopic tools, ultrasound) are less accurate, expensive or not available in every centre.
- **Shortage of specialized pathologists,** especially in low-resource settings.

4. The Scientific (AI) Problem

1. **Accurate tumor segmentation** in H&E images with a model that is fast enough for later real-time use and accurate enough for measurement.
2. **Scarce annotated data** and differences between laboratories (domain shift).
3. **Turning segmentation output into a quantitative measurement** in millimetres, not only a mask or a bounding box.
4. **Moving from public data to our own laboratory data** with little local data and limited computing resources.

Part III – Project Objectives

General objective

To build a **proof-of-concept** prototype that segments tumor regions in H&E histopathology images and estimates the distance between the tumor and the tissue boundary in millimetres, together with an honest evaluation of its accuracy and limitations, as a basis for later development toward intra-operative use.

Specific objectives

1. Train YOLO-seg models on the BCSS dataset, compare YOLO versions (v8, v10, v11) and select the best one.
2. Evaluate the segmentation with standard metrics (mask mAP, Dice, IoU).
3. Develop an algorithm that computes the shortest tumor-to-boundary distance in millimetres from the MPP calibration.
4. Integrate segmentation and measurement in one pipeline and measure the distance error (MAE/RMSE) and the processing time.
5. Test the model on microscope images from the university laboratory after improving them (Macenko normalization, downsampling), then adapt it by freezing early layers and fine-tuning.
6. Add a decision-support layer (RAG system or AI agent) as the end of the project.
7. Produce a scientific paper that is honest about its claims and limitations.

Part IV – State of the Art

> **Note**
>
> The descriptions below are high-level summaries of the reference list compiled for the project. The detailed inputs, outputs and reported figures of each study will be added in the next report.

Reference	Main idea	Relevance to our project
Shao et al., 2025 (World J Surg Oncol)	YOLOv11 for navigation of vacuum-assisted breast biopsy using ultrasound.	**Main comparison study:** a recent YOLO model in breast surgery, but on ultrasound images.
Veluponnar et al., 2023 (Cancers)	Margin assessment during breast surgery with a supporting intra-operative technique.	Clinical context of the margin problem.
Maris et al., 2025 (npj Breast Cancer)	Margin assessment with micro-PET-CT imaging.	An alternative technology for comparison.
Alhsnony & Sellami, 2024 (ICAART)	YOLOv8-seg applied to mammography images.	An example of YOLO-seg in breast imaging.
Frontiers in Surgery, 2025	YOLOv8 versus YOLOv12 in colon surgery.	Use of YOLO models in surgery at near real-time speed.
Amgad et al., 2019	Description of the BCSS dataset (breast cancer semantic segmentation).	Source of our training data and the way it was built.
Backup references	arXiv 2506.21006; a 2025 Frontiers in AI paper on oncology recommendations.	Additional support.

Research gap targeted by this project

Most of the works above operate on ultrasound, mammography, or spectroscopic and radiological techniques. **Segmentation of H&E histopathology images with YOLO-seg, followed by a quantitative distance measurement in millimetres,** is a direction that is less covered among the studies collected so far. We state this claim cautiously and will verify it with a more thorough literature review before the final version.

Part V – Proposed Solution

5. Overview

The system is developed in three stages over one academic year:

- **Semester 1:** a tumor-segmentation track (YOLO-seg trained on BCSS) and a distance-estimation track (on TCGA-BRCA whole-slide images), then integration and evaluation. BCSS was built from TCGA-BRCA slides, so the whole slides and the annotated regions come from the same source.
- **Semester 2:** laboratory experiments. Microscope images from university samples are improved and used to evaluate the model; then the **early layers are frozen** and the last layers are re-trained on laboratory data. The details of this mechanism will be studied and decided in this semester.
- **Final semester:** a decision-support layer, either a RAG system or an AI agent, as the end of the project.

6. Flow Chart of the System

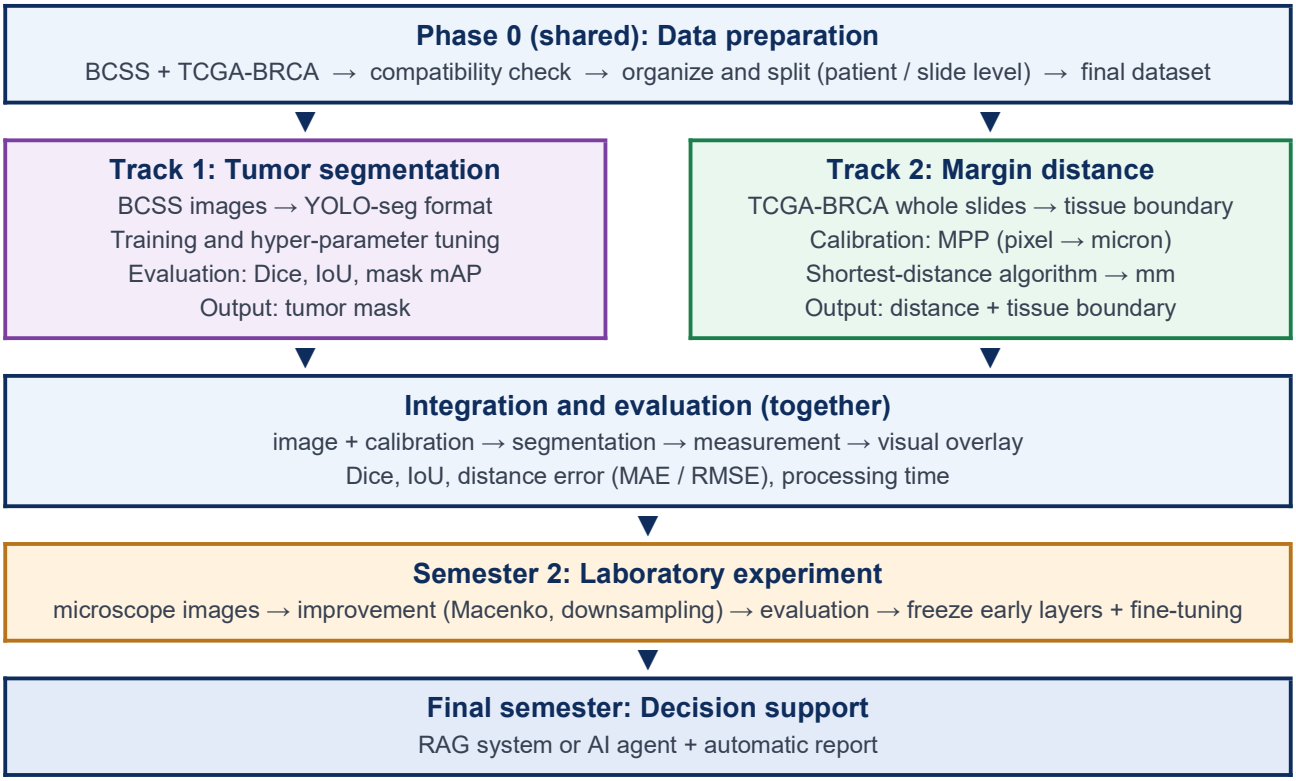


Figure 1. Block diagram of the proposed system and the three project stages.

7. Tentative Timeline for Semester 1

The following 8-week plan is an initial estimate and not a commitment.

Weeks	Activities
1–2	Check the datasets and annotations, confirm tissue-boundary availability, set up the environment and the data splits.
3–4	Parallel development: train YOLO-seg (segmentation track); build and test the distance algorithm (distance track).
5–6	Combine the outputs, run the end-to-end pipeline, fix problems and optimize.
7–8	Test on independent data, measure performance and errors, prepare the presentation and the report.

Part VI – Limitations and Honest Scope

To keep the work scientifically honest, we state the limitations explicitly. The project is a technical proof of concept.

Most important limitation

The distance we measure is between the tumor and the **edge of the tissue section on the slide**. This is a **proxy** for the surgical margin, because TCGA slides do not contain an annotated surgical ink margin. The results are therefore **not directly interpretable clinically**.

1. **Tissue boundaries are not annotated.** We generate them algorithmically (thresholding and clean-up) or manually for a small subset; their quality affects the measurement accuracy.
2. **Domain shift.** The model is trained on TCGA-derived data and may perform worse on our laboratory images despite Macenko normalization and layer freezing.
3. **Calibration.** Millimetre accuracy depends on a correct MPP; with the laboratory microscope a calibration slide or ruler is needed.
4. **Static images.** The current work operates on still images. "Real time" is a future goal and is not tested in the first semester.
5. **Laboratory samples** (including animal-tissue substitutes, if used) do not fully represent human tumors, and the amount of laboratory data is limited.
6. **Ethical and clinical framework.** The system is for research only and must not be used for treatment decisions; any use of human samples requires the necessary approvals.